

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EEE 311 — Past Questions & Materials
by Renance

Course: EEE 311
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 5
Levels: 200, 300

This study pack was compiled and restructured by Renance for EEE 311 students at Federal University of Petroleum Resources, Effurun. It brings together 5 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app
Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 5 — EEE 311 - 2024/2025 - First Semester - Circuit Theory 1

Level 200 • First Semester • PastQuestion

This source paper is a scanned document; its pages are reproduced below.

$\frac{3}{(s^2 + 8s + 16 + 2)}$
 $\frac{3}{(s+4)^2 + (\sqrt{2})^2}$

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

EXAMINATION: First Semester
 SESSION: 2024/2025
 DATE: 19/05/25



COURSE CODE: EEE 311
 COURSE TITLE: Circuit Theory I
 TIME ALLOWED: 3Hrs

INSTRUCTION: ANSWER FIVE QUESTIONS WITH ATLEAST TWO FROM EACH SECTION

SECTION A

Q1. The switch in the circuit of Fig. 1 is changed instantaneously from position "1" to "2". Given that $R_1 = 2\Omega$, $R_2 = 10\Omega$, $L = 0.1\text{H}$, $C = 0.8\text{mF}$ and $V_s = 120\text{V}$. Find the transient response of the inductive current.

14 marks

$\left(\frac{3}{\sqrt{2}} e^{-4t} \cdot \sin(\sqrt{2}t) \right) u(t)$

$z = B(1+2(-3))$
 $z = B(-5)$
 $c = -2/3$
 $-1/2$
 $z = B(3 + 1/2)$

at $t < 0$
 capacitor acts as open circuit

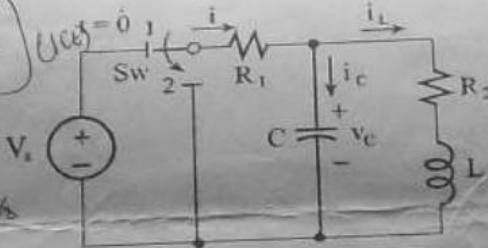


Fig. 1

Q2.(a) Define the following;

- Systems
- Electric Network
- Electric Circuit

3marks

(b) At what value of R will maximum power be dissipated in the circuit of Fig. 2

11marks

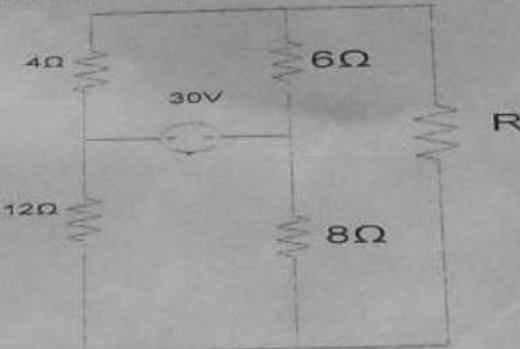


Fig. 2

Q3. (a) Find $v(t)$ in the circuit diagram of Fig. 3.

Bemarks

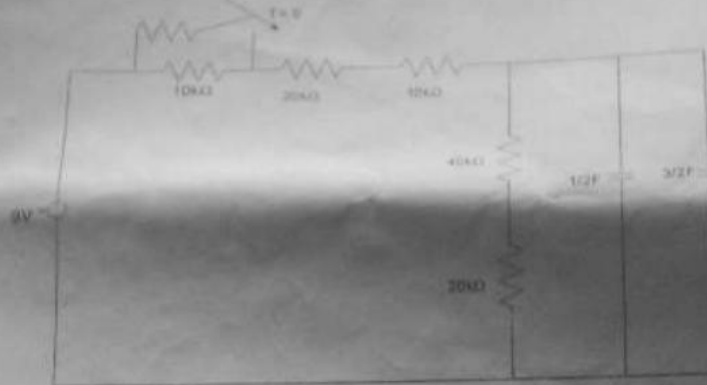


Fig. 3

(b) The ceiling fan in your bedroom was turn-on from 10pm till 6am. Note that, the coil in the fan was made of insulated copper wires. Use your knowledge of circuit theory to explain the condition of the coils before the fan was on during the period of 10:30pm and 5am. Use circuit diagrams for illustration if need be.

6marks

Q4. (a) Obtain the equivalent resistance R_{th} (R_{ab}) for the circuit in Fig. 4 and use it to find current I .

10 marks

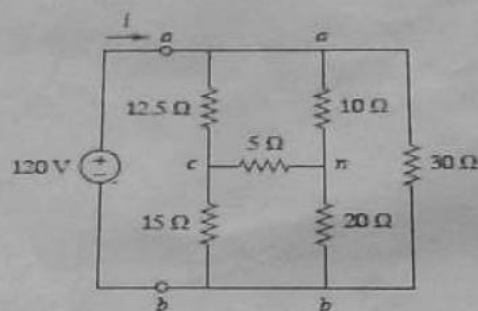


Fig. 4.

Handwritten calculations for equivalent resistance:

$$R_{th} = \frac{15 \times 20}{15 + 20} = \frac{300}{35} = 8.57 \Omega$$

$$R_{ab} = \frac{8.57 \times 30}{8.57 + 30} = \frac{257.1}{38.57} = 6.67 \Omega$$

13-48

(b) Define the following; (i) Time Constant (ii) Steady State Response.
(iii) What is the effect of time constant on the response of a circuit?

4marks

SECTION B

Q5. Find the trigonometrical fourier series for a full-wave rectified voltage sine waveform for the 2nd, 4th and 6th harmonics if the equation for the sinusoidal sine-waveform is given by $v(\theta) = V_m \sin \theta$ and the limit is $0 < \theta < \pi$ to $\pi < \theta < 2\pi$ (14 marks)

Q6 (a) Find the inverse fourier transform of the function

$$H(\omega) = \frac{3(3+2/j\omega)}{(1+j\omega)(4+j\omega)(2+j\omega)} \quad (7 \text{ marks})$$

Q6 (b) Find $v_o(t)$ in the circuit Figure 5. For $v_i(t) = 2e^{-3t} u(t)$. (7 marks)

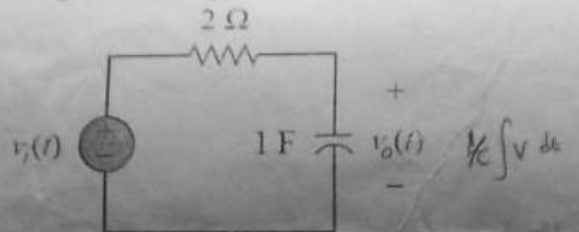


Figure 5

Q7. (a) Find the inverse laplace transform of the function

$$F(s) = \frac{5s^2 + 2}{s(s+1)(s+2)^2} \quad (7 \text{ marks})$$

(b) Find the voltage across the inductor $v_o(t)$ in the circuit diagram in Figure 6 assuming zero initial conditions. (7 marks)

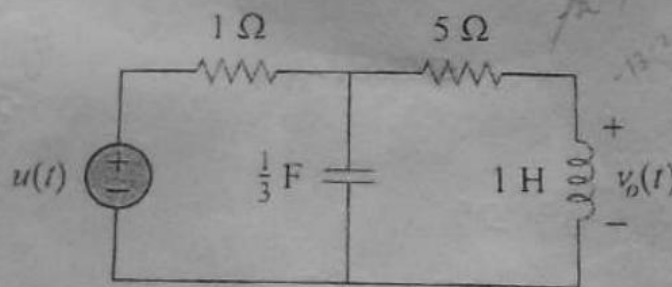


Figure 6.

Paper 2 of 5 — EEE 311 Circuit theory 2024/2025.

Level 300 • First Semester • PastQuestion

Aniknt® SAMSON = PROMUSE.

Coet [a4 20/2022

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF ENGINEERING TECHNOLOGY

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

EXAMINATION: First Semester

SESSION: 2024/2025 COURSE TITLE: Circuit Theory |

DATE: 19/05/25 TIME ALLOWED: 3Hrs

INSTRUCTION: ANSWER FIVE QUESTIONS WITH ATLEAST TWO FROM EACH SECTION

SECTION A

Q1. The switch in the circuit of Fig. 1 is changed instantaneously from position "1" to "2".

Given that $R_i = 20$, $R_o = 109$, $L = 0.1H$, $C = 0.8mF$ and $V_s = 120V$. Find the transient response of the inductive current. 14 marks

COURSE CODE: EEE 311

Q2.(a) Define the following;

(i) Systems

(ii) Electric Network

(iii) Electric Circuit 5 3marks

(b) At what value of R will maximum power be dissipated in the circuit of Fig. 2 11marks

es

42 +25 ze 60 ,

Q3. (a) Find $v(t)$ in the circuit diagram of Fig. 3. 8marks

\OKSU $T=0$

We

Laat

TOKS 20k 10k

40k

42F == /2F==

ov" :

20k2 s |

Fig. 3

(b) The ceiling fan in your bedroom was turn-on from 10pm till 6am. Note that, the coil in the fan was made of insulated copper wires. Use your knowledge of circuit theory to explain the condition of the coils before the fan was on, during the period of 10:30pm and 5am. Use circuit diagrams for illustration if need be. 6marks

Q4. (a) Obtain the equivalent resistance R_{tn} (R_{av}) for the circuit in Fig. 4 and use it to find

current |. 10 marks

i

—2 a

Sas 5100

sa L

120V © cL aw 3 s00

is L

sQ2 2

152 = = 202

b b

Fig. 4.

(b) Define the following; (i) Time Constant (ii) Steady State Response.

(iii) What is the effect of time constant on the response of a circuit? 4marks

ANKNO omE ek eee PEN LIA2O [28 22

SECTION B

Q5. Find the trigonometrical fourier series for a full-wave rectified voltage sine waveform for the 2", 4 and 6" harmonics if the equation for the sinusoidal sine-waveform is given by $f(t) = V_m \sin \omega t$ and the limit is $0 < \omega t < 2\pi$, (14 marks)

Q6.(a) Find the inverse fourier transform of the function

E_s

$H(\omega) = \frac{1}{(1 + j\omega)(4 + j\omega)(2 + j\omega)}$ (7 marks)

$(1 + j\omega)(4 + j\omega)(2 + j\omega)$

Q6.(b) Find $v_o(t)$ in the circuit Figure 5. For $v_i(t) = 2e^{-t} u(t)$. (7 marks)

20

MN

4

LF AR wu

Figure 5

Q7. (a) Find the inverse laplace transform of the function

$F(s) = \frac{5s+2}{s^2(s+1)(s+2)}$

7 k

$|s(s+1)(s+2)|^2$ a

(b) Find the voltage across the inductor $V_o(t)$ in the circuit diagram in Figure 6 assuming Zero initial conditions. (7 marks)

1a SQ.

ANN WV

$u(t) j\omega = IHS 4)$

Figure 6.

Paper 3 of 5 — EEE 311 Circuit Theory I section B 2025

Level 300 • First Semester • PastQuestion

SECTIONB ~

Q5. Find the trigonometrical fourier series for a full-wave rectified voltage sine waveform for the 274, 4 and 6 harmonics if the equation for the sinusoidal sine-waveform is given by $f(t) = V_m \sin \omega t$ and the limit is $0 < t < \pi$ to $\pi < t < 2\pi$. (14 marks)

Q6.(a) Find the inverse fourier transform of the function

$$3(34+2 \text{ Se})$$

$$H(\omega) = a \text{ ds (7 marks)}$$

$$(14 + j\omega)(44 + j\omega)(2 + j\omega)$$

Q6.(b) Find $v_o(t)$ in the circuit Figure 5. For $v_i(t) = 2e^{-t} u(t)$. (7 marks).

$v(t)$ IF AR lth fv ae

Figure 5 > al as

Q7. (a) Find the inverse laplace transform of the function $\frac{6}{s(s+1)(s+2)}$

(7 marks) :

$$\frac{6}{s(s+1)(s+2)}$$

Find the voltage across the inductor $v_o(t)$ in the circuit diagram in Figure 6 assuming

(b) Find the voltage across the inductor $v_o(t)$ in the circuit diagram in Figure 6 assuming

Zero initial conditions. (7 marks)

C 5 Q # 3 xi

x. DANS ie te?

AAAs NAN"] \$6 46" <

L+

oe 1H 2 4) "3 3

at ree a

oe

ins fe Fd

Figure 6. oe mee 4

Paper 4 of 5 — EEE 311 Circuit Theory I 2025

Level 300 • First Semester • PastQuestion

ne,

ob bo a et cal

oe 8S Pe Se

RCES, EFFURUN

U

FEDERAL UNIVERS! PETROLEUM RESO

COLLEGE O' GINEERING TECHNOLOGY te

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEER!

E 311

COURSE CODE: EE

5 ircuit Theory |

EXAMINATION: First Semester » 2

SESSION: 2024/2025 [-@ ' COURSE TITLE: ©

ts PAS TIME ALLOWED: 3Hrs

DATE: 19/05/25 Soe

INSTRUCTION: ANSWER FIVE QUESTIONS WITH ATLEAST TWO FROM EACH SECTION

"4" to "2".

taneously from position

=120V. Find the transient

44 marks

SECTION A

cuit of Fig. 1 is changed instan

Q1. The switch in the cir!

0.1H, C = 0.8mF and Vs

Given that Rs = 20, Re = 100, L=

response of the inductive current.

: az tx"

GRAS OS

2 BUTACY

22 8 (5) ra

< Ap ~lh

° a= ACs rH) Fig. 1

Q2.(a) Define the following; ;

(i) Systems a

(ii) Electric Network

(iii) Electric Circuit

(b) At what i i

value of R will maximum power be dissipated in the circuit of Fig. 2 11 k'

in . marks

3marks

= fe eae eg

Q3. (a) Find vit) in the circuit diagram of Fig. 3. 8marks

Nee

WWAr

eg | A pee aa ae

ee ee

40k rae 10K

i <

! 40k =

: ar = OF

aA

OVS t

i >

i

i

j 20k2 S

i

Fig. 3

| (b) The ceiling fan in your bedroom was turn-on from 10pm till 6am. Note that, the coil in the fan was made of insulated copper wires, Use your knowledge of circuit theory to explain the condition of the coils before the fan was on. during the nerind of 10:309m

end Sain. Use circuit diagrams for illustration if need be. 6marks

Q4. (a) Obtain the equivalent resistance R_{th} (R_{ab}) for the circuit in Fig. 4 and use it to find current i . 10 marks

i

celal: ee a

nso = = 100

; 52 R.

isn S = 22

Fs > rs

Fig. 4.

: 7 aa at

(b) Define the following; (i) Time Constant (ii) Steady State Response. i y

(iii) What is the effect of time constant on the response of a circuit? 4marks

Paper 5 of 5 — EEE 311 Test& Exam Solutions

Level 300 • First Semester • PastQuestion

This source paper is a scanned document; its pages are reproduced below.

Frequency Response

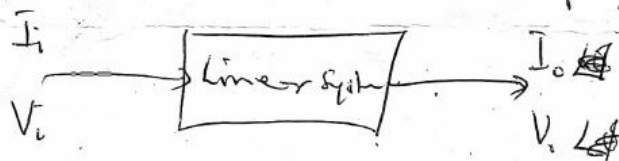
Frequency Response is the response of the circuit obtained when the amplitude of the sinusoidal source is kept constant and the frequency is varied.

Transfer function

This is an important tool in finding the frequency response of a circuit. It can be denoted as $H(\omega)$. It is defined as the frequency-dependent ratio of a phasor output $Y(\omega)$ to a phasor input $X(\omega)$.

$$H(\omega) = \frac{Y(\omega)}{X(\omega)}$$

Consider a linear system below



Using $N(\omega)$ to denote the numerator's value and $D(\omega)$ for the denominator's value, the frequency response of a circuit

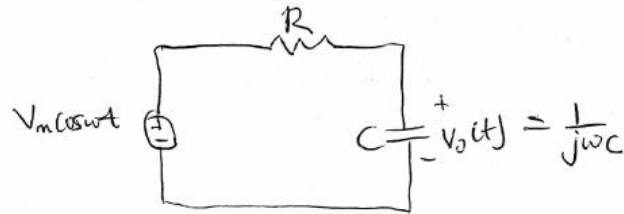
$$H(\omega) = \frac{N(\omega)}{D(\omega)}$$

It should however be noted that the roots of the numerator are called zeros. A zero is a value that results in a zero value of the function $H(\omega)$. While roots of denominator is called poles (ω).

Scanned with CamScanner

Example

For the circuit below, obtain the transfer function V_o/V_s and its frequency response.



$$H(\omega) = \frac{V_o}{V_s} = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \frac{1}{1 + j\omega RC}$$

$$|H(\omega)| = \frac{1}{\sqrt{1 + (\omega/\omega_0)^2}}, \quad \phi = -\tan^{-1} \omega/\omega_0$$

$$\text{where } \omega_0 = 1/RC$$

Bode Plots

Bode plots are semilog plots of the magnitude (in decibel) and phase (in degrees) of a transfer function versus frequency.

Examples

Construct the Bode plots for the transfer function

$$H(\omega) = \frac{200j\omega}{(j\omega + 2)(j\omega + 10)}$$

Solution

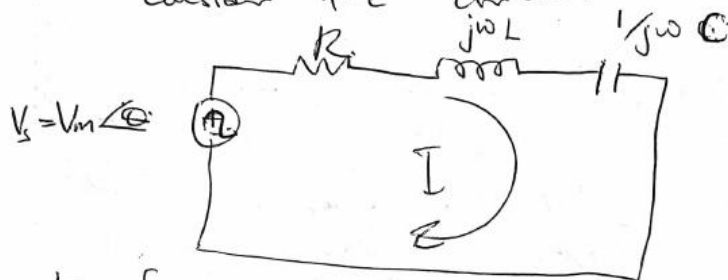
$$H(\omega) = \frac{10j\omega}{(1 + j\omega/2)(1 + j\omega/10)}$$

Scanned with CamScanner

Resonance

This is a condition in an RLC circuit which the capacitive and inductive reactances are equal in magnitude, thereby resulting in a purely resistive impedance. This phenomenon is very useful in the construction of filters.

Consider the circuit below.



In frequency domain, the input impedance is

$$Z = H(\omega) = \frac{V_s}{I} = R + j\omega L + \frac{1}{j\omega C}$$

$$Z = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

At resonance,

$$\omega L - \frac{1}{\omega C} = 0$$

For this condition to occur, the frequency that satisfies the occurrence is called resonant frequency, ω_0 .

$$\omega_0 L = \frac{1}{\omega_0 C}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/sec.}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}$$

The following are to be noted at resonance:

- The input impedance is purely resistive. (LC series combination acts like a short, and all the potential difference is across R).
- The source V_s and current I are in phase, so, power factor is unity.

Scanned with CamScanner

— The magnitude of the transfer function $H(\omega) = Z(\omega)$ is minimum.

— The V_L and V_C can be much more than V_S .

$$\text{i.e. } |V_L| = \frac{V_m}{R} \omega_0 L = Q V_m$$

$$|V_C| = \frac{V_m}{R} \frac{1}{\omega_0 C} = Q V_m$$

where Q is the quality factor

∴ The frequency response of the circuit's current magnitude

$$I = |I| = \frac{V_m}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}}$$

The average power dissipated by the RLC circuit is

$$P(\omega) = \frac{1}{2} I^2 R$$

The highest dissipated power occurs at resonance ($I = \frac{V_m}{R}$).

$$P(\omega_0) = \frac{1}{2} \frac{V_m^2}{R}$$

The "sharpness" of the resonance in a resonant circuit is measured quantitatively by the quality factor Q .

$$Q = 2\pi \frac{\text{Peak energy stored in the circuit}}{\text{Energy dissipated by the circuit in one period of resonance.}}$$

In short,

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R}$$

Bandwidth B is defined as the difference between the half-power frequencies

$$\text{i.e. } B = \omega_2 - \omega_1$$

The relationship between the Q -factor and the bandwidth is given as

$$B = \frac{R}{L} = \frac{\omega_0}{Q} \quad \text{only for series RLC}$$

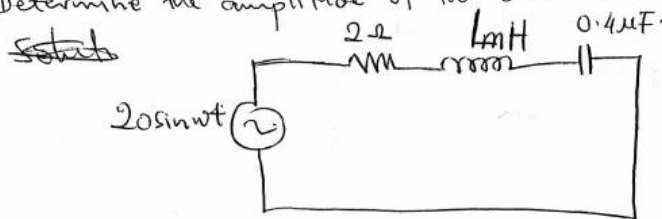
In other words, the quality factor Q of a resonant circuit can be defined as the ratio of its resonant frequency

Scanned with CamScanner

to its bandwidth.

Example

In the circuit below, find (i) the resonant frequency and the half-power frequencies. (ii) Quality factor and bandwidth. (iii) Determine the amplitude of the current at ω_0 , ω_1 , and ω_2 .



Solution

(i) The resonant frequency is given as

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.3 \times 0.4 \times 10^{-6}}} = 50 \text{ krad/s}$$

To find ω_1 (lower half power frequency)

$$\omega_1 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} = 49 \text{ krad/s}$$

ω_2 (upper half power frequency).

$$\omega_2 = 1 + \sqrt{1 + 2500} \text{ krad/s} = 51 \text{ krad/s}$$

(ii) $B = \omega_2 - \omega_1 = 2 \text{ krad/s}$

$$Q = \frac{\omega_0}{B} = \frac{50}{2} = 25$$

(iii) At ω_0

$$I = \frac{V_m}{R} = \frac{20}{2} = 10 \text{ A}$$

At ω_1, ω_2

$$I = \frac{V_m}{\sqrt{2} R} = \frac{10}{\sqrt{2}} = 7.07 \text{ A}$$

Scanned with CamScanner

Note

For parallel resonance,

$$B = \omega_c - \omega_1 > \frac{1}{R_c}$$

$$\text{and } Q = \frac{R}{\omega_0 L} = \omega_0 R C$$

$$\omega_1 = -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{Lc}} \quad \text{or } \omega_0 \sqrt{1 + \left(\frac{1}{2Q}\right)^2} - \frac{\omega_0}{2Q}$$

$$\omega_2 = \frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{Lc}} \quad \text{or } \omega_0 \sqrt{1 + \left(\frac{1}{2Q}\right)^2} + \frac{\omega_0}{2Q}$$

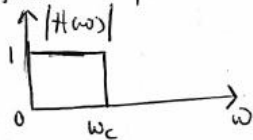
Filters

A filter is a circuit designed to pass signals with desired frequencies and reject or attenuate others.

A passive filter is defined as that filter containing only passive elements (RLC). It is active, if it consists of the active elements (transistors and op amps), in addition to the passive elements.

Types of filters

- 1.) Lowpass filter: passes low frequencies and stops high frequencies.

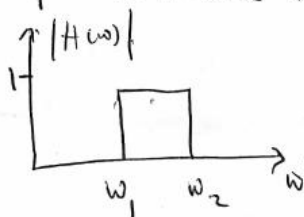


- 2.) Highpass filter: passes high frequencies and stops low frequencies.

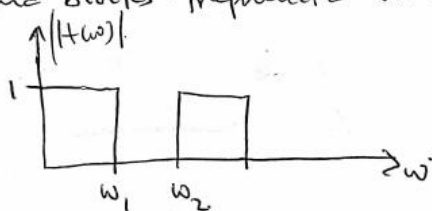


Scanned with CamScanner

~~Bandpass~~ filter: passes frequencies within a frequency band and blocks frequencies outside the band.

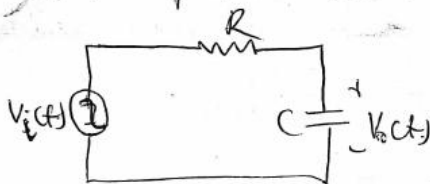


4.) Bandstop filter: passes frequencies outside a frequency band and blocks frequencies within the band.



Lowpass filters

A low pass filter can be formed when the output of RC circuit is taken off the capacitor or RL circuit off the resistor.



$$H(\omega) = \frac{V_o}{V_i} = \frac{1/j\omega C}{R + 1/j\omega C}$$

$$H(\omega) = \frac{1}{1 + j\omega RC}$$

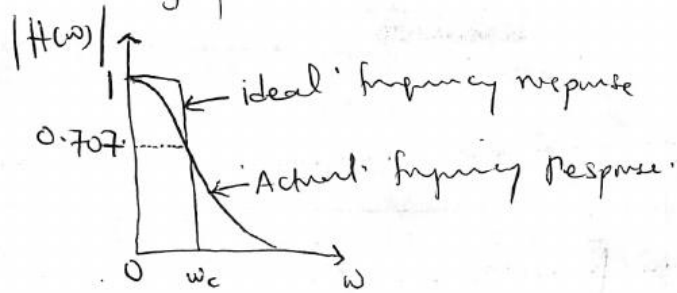
The half power frequency which is equivalent to the corner frequency called cutoff frequency ω_c is obtained by,

$$|H(\omega_c)| = \frac{1}{\sqrt{1 + \omega_c^2 R^2 C^2}} = \frac{1}{\sqrt{2}}$$

$$\boxed{\omega_c = \frac{1}{RC}}$$

Scanned with CamScanner

The X-axis graph is drawn as:

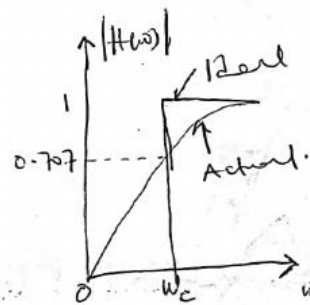


Highpass Filters:

$$H(\omega) = \frac{V_o}{V_i} = \frac{R}{R + 1/j\omega C}$$

$$H(\omega) = \frac{j\omega RC}{1 + j\omega RC}$$

$$\omega_c = \frac{1}{RC}$$



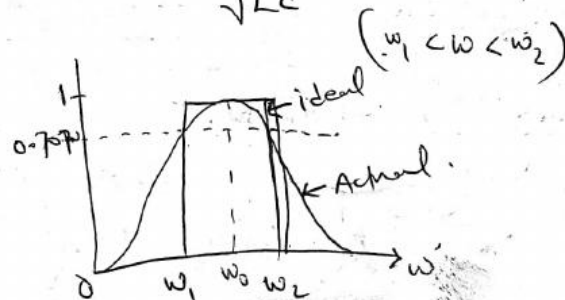
Bandpass filter

The RLC series resonant circuit provides a bandpass filter when the output is taken off the resistor.

$$H(\omega) = \frac{V_o}{V_i} = \frac{R}{R + j(\omega L - 1/\omega C)}$$

The centre frequency is given as:

$$\omega_0 = \frac{1}{\sqrt{LC}}$$



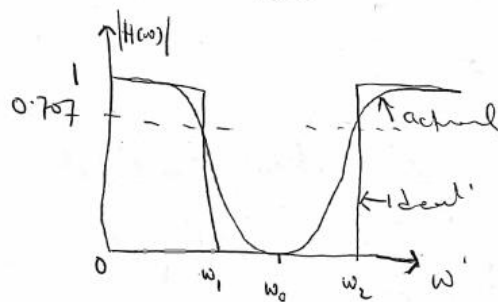
Scanned with CamScanner

Bandstop Filter

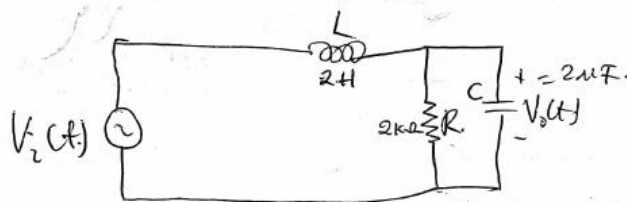
Bandstop filter is formed when the output of RLC series resonant circuit is taken off the LC series combination.

$$H(\omega) = \frac{V_o}{V_i} = \frac{j(\omega L - 1/\omega C)}{R + j(\omega L - 1/\omega C)}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$



Example



For the circuit above, determine what type of filter is shown. Also, calculate the corner or cut off frequency.

Solution

$$H(s) = \frac{V_o}{V_i} = \frac{R // 1/sC}{sL + R // 1/sC}$$

$$R // \frac{1}{sC} = \frac{R/sC}{R + 1/sC} = \frac{R}{1 + sRC}$$

$$H(s) = \frac{R/(1 + sRC)}{sL + R/(1 + sRC)} = \frac{R}{s^2RLC + sL + R}$$

Scanned with CamScanner

$$|H(s)| = \frac{R}{\sqrt{(R - \omega^2 LC)^2 + \omega^2 L^2}}$$

At corner frequency $|H(s)|^2 = \frac{1}{2}$

$$|H(s)|^2 = \frac{1}{2} = \frac{R^2}{(R - \omega_c^2 LC)^2 + \omega_c^2 L^2}$$

$$2 = (1 - \omega_c^2 LC)^2 + \left(\frac{\omega_c L}{R}\right)^2$$

$$2 = (1 - 4\omega_c)^2 + \omega_c^2$$

$$16\omega_c^4 - 7\omega_c^2 - 1 = 0$$

$$\therefore \omega_c^2 = 0.5509$$

$$\omega_c = 0.742 \text{ krad/s} = 742 \text{ rad/s}$$

Scanned with CamScanner

Fourier Transform and its application to circuit.

The Fourier transform of a function $f(t)$ is given as

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

It is in general a complex function.

$$F(\omega) = \text{Re}\{F(\omega)\} + j \text{Im}\{F(\omega)\} = |F(\omega)| e^{j\phi(\omega)}$$

The inverse Fourier transform is defined as

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

Properties and Theorems of Fourier Transform.

1.) Linearity.

$$a_1 f_1(t) + a_2 f_2(t) + \dots + a_n f_n(t) \Leftrightarrow a_1 F_1(\omega) + a_2 F_2(\omega) + \dots + a_n F_n(\omega)$$

(2.) Symmetry.

If $F(\omega)$ is $F\{f(t)\}$, then

$$F(t) \Leftrightarrow 2\pi f(-\omega)$$

$$\text{i.e. } f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

$$2\pi f(-t) = \int_{-\infty}^{\infty} F(\omega) e^{-j\omega t} d\omega$$

interchanging t and ω ,

$$2\pi f(\omega) = \int_{-\infty}^{\infty} F(t) e^{-j\omega t} dt$$

3.) Time Scaling.

$$f(at) \Leftrightarrow \frac{1}{|a|} F\left(\frac{\omega}{a}\right)$$

Scanned with CamScanner

4.) Time shifting

$$f(t-t_0) \Leftrightarrow F(\omega) e^{-j\omega t_0}$$

5.) Frequency shifting

$$e^{j\omega_0 t} f(t) \Leftrightarrow F(\omega - \omega_0)$$

6.) Time Differentiation

$$\frac{d^n}{dt^n} f(t) \Leftrightarrow (j\omega)^n F(\omega)$$

7.) Frequency Differentiation

$$(-jt)^n f(t) \Leftrightarrow \frac{d^n}{d\omega^n} F(\omega)$$

8.) Time Integration

$$\int_{-\infty}^t f(\tau) d\tau \Leftrightarrow \frac{F(\omega)}{j\omega} + \pi F(\omega) \delta(\omega)$$

9.) Conjugate Time and Frequency Functions

$$f^*(t) \Leftrightarrow F^*(-\omega)$$

10.) Time Convolution

$$f_1(t) * f_2(t) \Leftrightarrow F_1(\omega) F_2(\omega)$$

11.) Frequency Convolution

$$f_1(t) f_2(t) \Leftrightarrow \frac{1}{2\pi} F_1(\omega) * F_2(\omega)$$

12.) Area under $f(t)$

$$F(0) = \int_{-\infty}^{\infty} f(t) dt$$

Scanned with CamScanner

(13.) Area under $F(\omega)$.

$$f(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) d\omega.$$

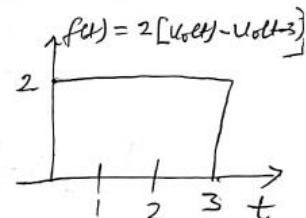
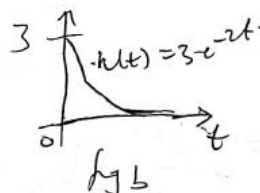
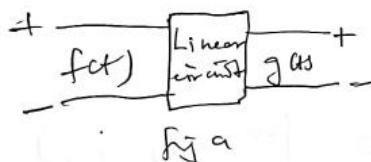
(14.) Parseval's Theorem.

$$\int_{-\infty}^{\infty} |f(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |F(\omega)|^2 d\omega.$$

$$g(t) = f(t) * h(t)$$

Application to circuit

For the linear circuit below, Use Fourier transform method to compute the response $g(t)$ when the input $f(t)$ is as shown in fig c.



Solution

The input $f(t) = f_1(t) + f_2(t)$

$$f_1(t) = 2u_0(t) \text{ and } f_2(t) = -2u_0(t-3)$$

$H(\omega)$ is the Fourier transform of $h(t)$

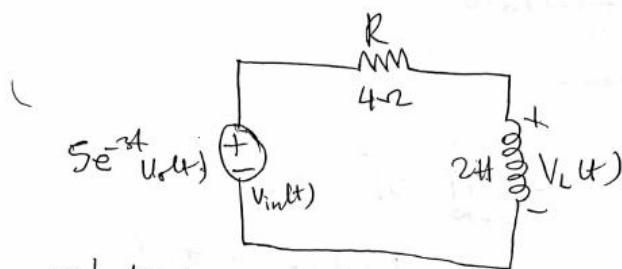
$$F\{h(t)\} = H(\omega) = \frac{3}{j\omega + 2}$$

$$F_1(\omega) = 2\left(\frac{1}{j\omega} + \frac{1}{j\omega}\right)$$

$$\therefore G_1(\omega) = H(\omega) \cdot F_1(\omega) = \frac{3}{j\omega + 2} \cdot 2\left(\frac{1}{j\omega} + \frac{1}{j\omega}\right)$$

Scanned with CamScanner

2.) For the circuit below, use power transfer method and system function $H(\omega)$ to find $V_L(t)$. Assume $i_L(0^-) = 0$.



Solution.

The phasor equivalent circuit is drawn as in



$$V_{out}(\omega) = \frac{j2\omega}{4 + j2\omega} V_{in}(\omega) = \frac{j\omega}{j\omega + 2} V_{in}(\omega)$$

The system function $H(\omega)$ is

$$H(\omega) = \frac{V_{out}(\omega)}{V_{in}(\omega)} = \frac{j\omega}{j\omega + 2}$$

$$V_{in}(t) = 5e^{-2t} \text{ volt} \Rightarrow V_{in}(\omega) = \frac{5}{j\omega + 3}$$

$$\therefore V_{out}(\omega) = H(\omega) V_{in}(\omega) = \frac{j\omega}{j\omega + 2} \cdot \frac{5}{j\omega + 3}$$

Using partial fraction expansion.

$$\frac{5j\omega}{(j\omega + 2)(j\omega + 3)} = \frac{A}{j\omega + 2} + \frac{B}{j\omega + 3}$$

Scanned with CamScanner

$$A = -10 \quad \text{and} \quad B = 15$$

$$\hat{V}_{out}(s) = \frac{15}{j\omega + 3} - \frac{10}{j\omega + 2}$$

$$V_L(t) = V_{out}(t) = \mathcal{F}^{-1} \left\{ \frac{15}{j\omega + 3} + \frac{-10}{j\omega + 2} \right\}$$

$$= 15e^{-3t} - 10e^{-2t}$$

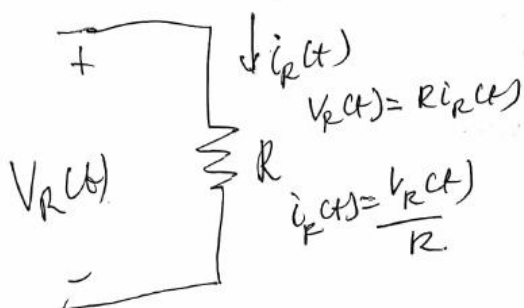
$$V_L(t) = 5(3e^{-3t} - 2e^{-2t})u_0(t)$$

Scanned with CamScanner

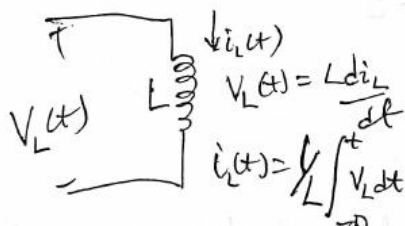
①

Circuit Analysis with Laplace Transform

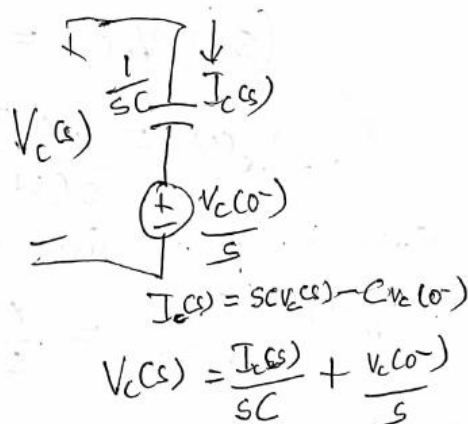
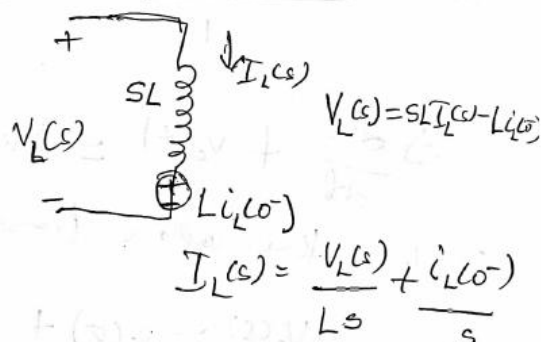
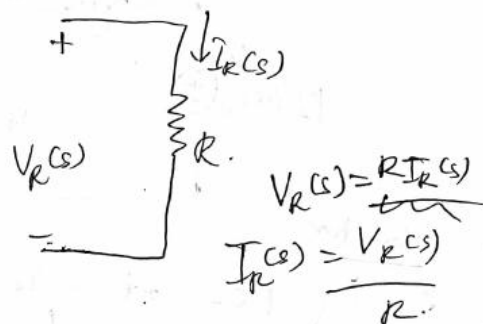
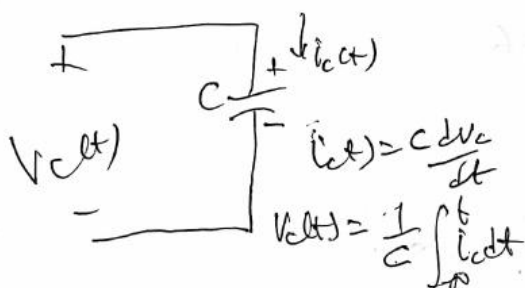
At first, let us see the transformation of the basic passive elements from time domain to complex frequency domain.



For inductor



For capacitor

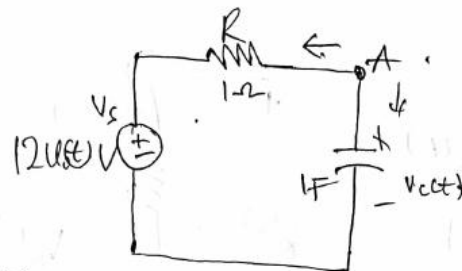


Scanned with CamScanner

2)

Example

Use Laplace transform method to obtain the response of the circuit, given that $v_c(0^-) = 6V$.

Solution

Applying KCL at node A

$$i_R + i_C = 0$$

$$\Rightarrow \frac{v_c(t) - 12u(t)}{1} + 1 \cdot \frac{dv_c}{dt} = 0$$

$$\Rightarrow \frac{dv_c}{dt} + v_c(t) = 12u(t)$$

Taking the Laplace transform of the above equation,

$$sV_c(s) - v_c(0^-) + V_c(s) = \frac{12}{s}$$

$$(s+1)V_c(s) = \frac{12}{s} + 6$$

$$\therefore V_c(s) = \frac{6s+12}{s(s+1)}$$

Going by partial fraction expansion.

$$V_c(s) = \frac{6s+12}{s(s+1)} = \frac{A}{s} + \frac{B}{s+1}$$

Scanned with CamScanner

$$A = 12 \text{ or } B = -6$$

$$\therefore V_C(s) = \frac{12}{s} - \frac{6}{s+1} \Rightarrow 12 - 6e^{-t}$$

$$= (12 - 6e^{-t}) u_0(t) = V_C(t)$$

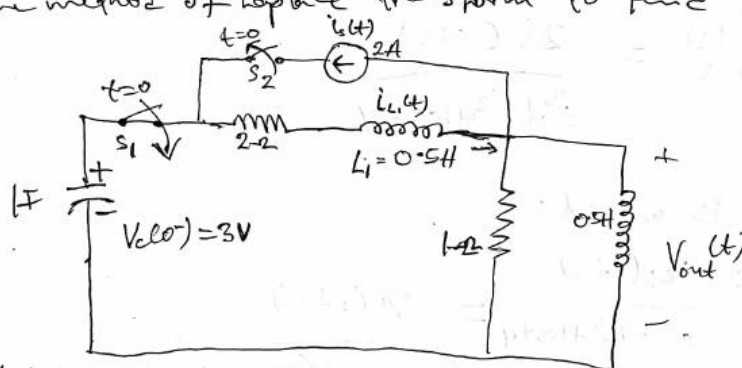
Also using KVL

$$R i_C(t) + \frac{1}{C} \int_{-\infty}^t i_C(t) dt = 12 u_0(t)$$

$$i_C(t) + \int_{-\infty}^t i_C(t) dt = 12 u_0(t)$$

Example 2

In the ckt below, S_1 closes at $t=0$, while S_2 opens. Use the method of Laplace transform to find $V_{out}(t)$ for $t > 0$



Solution

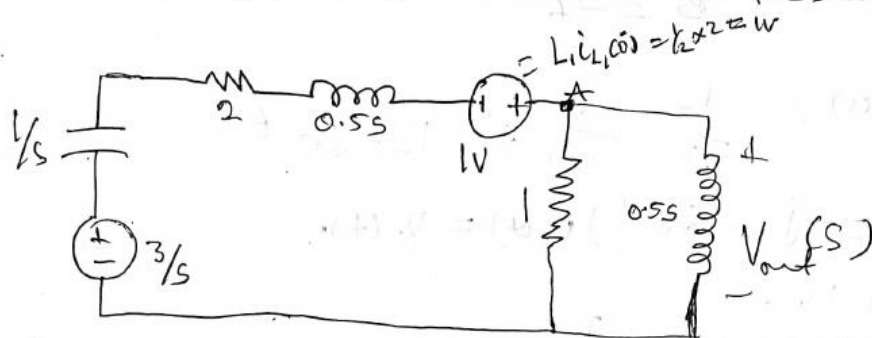
The circuit contains both inductor and capacitor, so we need two initial conditions.

1st one is $V_C(0^-) = 3V$ and the other initial condition is across inductor L_1

$$\therefore i_{L_1}(0^-) = 2A$$

Scanned with CamScanner

Now for $t > 0$, we break the circuit into the s-domain.



Apply KCL at node A

$$\frac{V_{out}(s) - 1 - \frac{3}{s}}{\frac{1}{s} + 2 + \frac{s}{2}} + \frac{V_{out}(s)}{1} + \frac{V_{out}(s)}{\frac{s}{2}} = 0$$

Simplifying

$$V_{out}(s) = \frac{2s(s+3)}{s^3 + 8s^2 + 10s + 4}$$

-6.5708 is a root.

$$\therefore V_{out}(s) = \frac{2s(s+3)}{s^3 + 8s^2 + 10s + 4} = \frac{2s(s+3)}{(s+6.57)(s^2 + 1.43s + 0.61)}$$

Using partial fraction expansion.

$$V_{out}(s) = \frac{2s(s+3)}{(s+6.57)(s^2 + 1.43s + 0.61)} = \frac{A}{s+6.57} + \frac{Bs+C}{s^2 + 1.43s + 0.61}$$

$$A = 1.36$$

Scanned with CamScanner

$$r_2 = 0.64$$

$$r_3 = -0.12$$

$$\therefore V_{out}(s) = \frac{1.36}{s+6.57} + \frac{0.64s - 0.12}{s^2 + 1.43s + 0.61}$$

Using algebra:

$$\Rightarrow \frac{1.36}{s+6.57} + \frac{0.64s + 0.46 - 0.58}{s^2 + 1.43s + 0.51 + 0.1}$$

$$V_{out}(s) = \frac{1.36}{s+6.57} + \frac{(0.64)s + 0.75 - 0.9}{(s+0.715)^2 + (0.316)^2}$$

$$= \frac{1.36}{s+6.57} + \frac{0.64(s+0.715)}{(s+0.715)^2 + (0.316)^2} - \frac{0.58}{(s+0.715)^2 + (0.316)^2}$$

$$= \frac{1.36}{s+6.57} + \frac{0.64(s+0.715)}{(s+0.715)^2 + (0.316)^2} - \frac{1.84 \times 0.316}{(s+0.715)^2 + (0.316)^2} \quad (6)$$

(Inverse Laplace Transform)

$$V_{out}(t) = (1.36e^{-6.57t} + 0.64e^{-0.715t} \cos 0.316t - 1.84e^{-0.715t} \sin 0.316t) u(t)$$

Scanned with CamScanner